



COMPETENCY STANDARD
FOR
MAINTENANCE OF JUTE MILLS MACHINERY
(JUTE SECTOR)

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Maintenance of Jute Mills Machinery (Jute Sector)** is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

The document was developed under the Skills for Employment Investment Program (SEIP) and subsequently reviewed and updated with the engagement of occupation-specific industry experts to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been reviewed and updated by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-MJM-01-G	Apply occupational health and safety (OHS) practices at workplace	1. Identify, control and report OHS hazards 2. Conduct work safely 3. Follow emergency response procedures 4. Maintain and improve health and safety in the workplace	10
SICIP- JUTE-MJM-02-G	Work In a Team Environment	1. Identify team goals and work processes 2. Communicate and co-operate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hour			20

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-MJM-01-S	Apply Quality Procedures in Jute Manufacturing	1. Identify quality procedures 2. Follow quality procedures 3. Maintain standard procedures	10
SICIP- JUTE-MJM-02-S	Apply green practices in Jute sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			20

Occupation Specific Competencies (180 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-JUTE-MJM-01-O	Interpret Jute Mills Machinery	1. Identify jute mill machinery and their functions. 2. Explain the operating principles of jute mill machinery. 3. Identify major machine components and their functions. 4. Interpret machine operating parameters and performance requirements. 5. Identify safety requirements and operational hazards associated with jute mill machinery.	30
SICIP-JUTE-MJM-02-O	Perform Preventive Maintenance of Jute Machinery	1. Plan and prepare for preventive maintenance 2. Carry out routine inspection of machinery 3. Perform cleaning, oiling, greasing and lubrication of machine parts 4. Perform minor adjustment and tightening 5. Complete and report maintenance activities	30
SICIP-JUTE-MJM-03-O	Diagnose Mechanical Faults in Jute Machinery	1. Identify symptoms of machine malfunction 2. Use diagnostic tools and techniques 3. Analyze root cause of faults 4. Determine and recommend corrective action 5. Document fault diagnosis findings Interpret international rules and regulations	30
SICIP-JUTE-MJM-04-O	Carry Out Repair and Replacement of Jute Machinery Components	1. Prepare for repair/replacement work 2. Dismantle machine components 3. Repair or replace defective	30

Code	Unit of Competency	Elements of Competency	Duration (hours)
		components 4. Reassemble and reinstall components 5. Perform testing and commission repaired machine	
SICIP-JUTE-MJM-05-O	Perform Machine Testing and Commissioning	1. Prepare machine for alignment and calibration 2. Perform mechanical alignment 3. Calibrate machine settings 4. Commission and validate machine performance	25
SICIP-JUTE-MJM-06-O	Maintain Jute Processing Machines	1. Maintain batching and softening machines. 2. Maintain breaker card and finisher card machines 3. Maintain drawing frame and spinning frame machines. 4. Maintain winding, beaming, and weaving machines. 5. Maintain finishing and baling equipment. 6. Verify machine performance and production readiness after maintenance.	35
Total			180

COMPETENCY STANDARD: MAINTENANCE OF JUTE MILLS MACHINERY

Generic Competencies:

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES AT WORKPLACE	Nominal Duration: 10 Hrs.	Unit Code: SICIP-JUTE-MJM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices at workplace. It specifically includes the tasks of identifying, controlling and reporting OHS hazards, conducting work safely, following emergency response procedures and maintaining and improving health and safety in the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify, control and report OHS hazards	1.1 Work area is routinely checked for Occupational Health and Safety (OHS) hazards prior to commencing and during work. 1.2 <u>Hazards</u> are identified and corrective action is taken within the level of responsibility. 1.3 OSH hazards and incidents are reported to appropriate personnel according to workplace procedures. 1.4 <u>Safety signs and symbols</u> are identified and followed. 1.5 <u>Preventive Control measures</u> are identified in accordance with OSH work standards.
2. Conduct work safely	2.1 OHS practices are applied in the workplace. 2.2 <u>Personal Protective Equipment (PPE)</u> are used.
3. Follow emergency response procedures	3.1 Emergency situations are identified and reported according to workplace requirements. 3.2 Emergency procedures are followed as appropriate to the nature of the emergency and according to workplace procedures. 3.3 <u>Workplace procedures</u> for dealing with accidents, fires and emergencies are followed whenever necessary within scope of responsibilities.
4. Maintain and improve health and safety in the workplace	4.1 Risks are identified and appropriate control measures are implemented in the workplace. 4.2 Recommendations arising from risk assessments are

	implemented within level of responsibility.
	4.3 Safety records are maintained according to <u>company policies.</u>

Range of Variables

Variable	Range (May include but not limited to):
1. Hazards	1.1. OHS incidents include near misses, injuries, illnesses and property damage, noise, handling hazardous substances. 1.2. Biological hazards- bacteria, viruses, plants, parasites, mites, molds, fungi, insects 1.3. Chemical hazards – dusts, fibers, mists, fumes, smoke, gasses, vapors 1.4. Ergonomics Hazards 1.5. Physiological factors – monotony, personal relationship, work out cycle 1.6. Safety hazards (unsafe workplace condition) – confined space, excavations, falling objects, gas leaks, electrical, poor storage of materials and waste, spillage, waste and debris 1.7. Unsafe workers' act (Smoking in off-limited areas, Substance and alcohol abuse at work) 1.8. Working with and near moving equipment / load shifting equipment. 1.9. Broken or damaged equipment or materials.
2. Safety signs and symbols	2.1 Direction signs (exit, emergency exit, etc.) 2.2 First aid signs 2.3 Danger Tags 2.4 Hazard signs 2.5 Safety tags 2.6 Warning signs
3. Personal Protective Equipment (PPE)	3.1 Apron 3.2 Safety Helmet 3.3 Goggles 3.4 Ear muffs 3.5 Ear plugs 3.6 Gloves 3.7 Clothing 3.8 Safety Boots 3.9 Face Protection (Musk & Shield) 3.10 Scarf 3.11 Hair Net/cap/bonnet

	3.12 Hard hat
4. Preventive Control Measures	4.1 Eliminate the hazard (i.e., get rid of the dangerous machine) 4.2 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 4.3 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 4.4 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OHS-related topics, issue warning signage, rotation/shifting work schedule) 4.5 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 4.6 Use personal protective equipment 4.7 Safety, health and work environment evaluation 4.8 Periodic and/or special medical examinations of workers
5. Workplace Procedures	5.1 Fire fighting 5.2 Medical and first aid 5.3 Evacuation 5.4 Material Safety Data Sheets (MSDSs) and manufacturers' advice
6. Company policies	6.1 Job related Standard Operating Procedures (SOPs). 6.2 Occupational Health and Safety (OHS) specific procedures. 6.3 Examples of OHS procedures include – consultation and participation, emergency response to specific hazards, incident investigation, risk assessment, reporting arrangement and issue resolution procedures.

Curricular Content Guide

1. Underpinning Knowledge	1.1. OHS Workplace Policies and Procedures. 1.2. Work safety procedures. 1.3. Fire and emergency procedures. 1.4. Types of Hazards (Biological, Chemical and Physical) and their effects. 1.5. PPE types and uses. 1.6. Personal hygiene practices. 1.7. OHS awareness. 1.8. Steps of hazard identification. 1.9. Principles of hazards control. 1.10. Employer's role.
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	1.11. Supervisor's responsibilities.
2. Underpinning Skills	2.1. Identifying OHS policies and procedures. 2.2. Following personal work safety practices. 2.3. Reporting hazards and risks. 2.4. Responding to emergency procedures. 2.5. Maintaining physical well-being in the workplace. 2.6. Identify tools and equipment related to OHS. 2.7. Improving OHS performance.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Manuals 4.4 Workplace documents, signs and symbols 4.5 Relevant specifications or work instructions. 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 used Personal Protective Equipment (PPE). 1.2 identified hazards. 1.3 took corrective action of different hazards. 1.4 took corrective action for emergency procedure. 1.5 reported emergency situation to the Supervisor / Manager. 1.6 satisfied requirements mentioned in the performance criteria and range of variables.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated

	work place after completion of the training.
	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-JUTE-MJM-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work in a team environment. It specifically includes the tasks of Identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes.	1.1. Team goals and collaborative decision-making processes are identified. 1.2. Role and common goals of the team are defined from available <u>sources of information</u> . 1.3. Team structure, responsibilities and reporting relations are identified from team discussions and other external sources.
2. Communicate and co-operate with team members.	2.1 Communication and negotiation skills are applied and maintained in all relevant situations. 2.2 Constructive contributions are made to <u>workplace discussions</u> on such issues as production, quality and safety. 2.3 Goals/ objectives and action plans undertaken in the workplace are communicated promptly. 2.4 Information regarding problems and issues are organized coherently to ensure clear and effective communication. 2.5 Dialogue is initiated with appropriate personnel. 2.6 Communication problems and issues are raised 2.7 Barriers to communication are identified and resolved
3. Work as a team member	3.1 Effective forms of communication are used to interact with <u>team members</u> in discussing team activities and objectives. 3.2 Mutual respect, empathy and active collaboration are

	demonstrated 3.3 Communication channels are followed as per <u>workplace context.</u>
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified 4.2 Problems are investigated and analyzed 4.3 Potential solutions of problem are identified 4.4 Recommendations about possible solutions are developed, documented, ranked and presented to team members for decision.

Range of Variables

Variable	Range (May include but not limited to):
1. Sources of information	1.1. Organizational structures 1.2. Operations Manuals 1.3. Job description 1.4. Standard operating procedures
2. Workplace discussions	2.1 Coordination meetings 2.2 Toolbox discussion 2.3 Peer-to-peer discussion
3. Team members	3.1 Coach / members 3.2 Supervisor / manager 3.3 Peers / colleagues 3.4 Other members /Employee representative of the organization.
4. Workplace context	4.1 National laws and statutes 4.2 Standard operating procedures 4.3 Workplace rules and regulations

Curricular Content Guide

1. Underpinning Knowledge	1.1. Sources of information define 1.2. Team structure, role, and responsibility. 1.3. Individual member's roles and responsibilities. 1.4. Effective verbal communication methods 1.5. Communication flow and reporting structures. 1.6. Interpersonal communication skills. 1.7. Organization requirements for written and electronic communication methods 1.8. Communication problems and issues 1.9. Barriers in communication 1.10. Team planning. 1.11. Team meeting procedures. 1.12. Workplace etiquette
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	1.13. Industry maintenance, service and helpdesk practices, processes and procedures 1.14. Industry standard diagnostic tools 1.15. Malfunctions and resolutions
2. Underpinning Skills	2.1 Organizing sources of information 2.2 Identifying the role and responsibility of the team. 2.3 Identifying roles and responsibilities of individual members. 2.4 Identifying effective verbal communication methods 2.5 Identifying communication flow and reporting structure. 2.6 Identifying interpersonal communication skills 2.7 Complying with organization requirements for the use of written and electronic communication methods 2.8 Negotiation and communication skills 2.9 Participating in team discussion. 2.10 Working as a team member. 2.11 Participating in a variety of workplace discussions 2.12 Effective clarifying and probing skills 2.13 Identifying issues 2.14 Identifying current industry standard diagnostic tools 2.15 Describing common malfunctions and resolutions. 2.16 Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Relevant specifications or work instructions. 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 identified own role and responsibilities within team 1.1 communicated and cooperated with team members
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	1.2 practice problem solving within the team
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Sector Specific Competencies (20 Hrs.)

Unit of Competency: APPLY QUALITY PROCEDURES IN JUTE MANUFACTURING	Nominal Duration: 10 Hrs.	Unit Code: SICIP-JUTE-MJM-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply quality procedures in jute manufacturing. It specifically includes the tasks of identifying quality procedures, following quality procedures and maintaining standard procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify quality procedures	1.1. <u>Manuals</u> and guidelines for jute production processes are collected as per product specifications. 1.2. The importance of manuals in ensuring quality standards is recognized. 1.3. Instructions and procedures for quality control are identified. 1.4. Required information is collected from manuals. 1.5. Performance measurement systems are identified.
2. Follow quality procedures	2.1 Instructions and procedures are followed strictly and duties are performed in accordance with the <u>quality improvement system</u> . 2.2 Concept of supplying jute products to meet <u>customer quality requirements</u> is understood and applied. 2.3 Conformance to specifications is ensured. 2.4 Defects and deviations are detected and reported to the relevant authority according to standard operating procedures.
3. Maintain standard procedures	3.1 Performance is assessed at regular intervals. 3.2 Specifications and standard operating procedures are established. 3.3 Quality of jute product is checked and verified. 3.4 Quality control and quality assurance system procedures for each job are followed. 3.5 Conformance to specification is ensured in all situations.

Range of Variables

Variable	Range (May include but not limited to):
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1. Manuals	1.1 Buyers' specification manual 1.2 Compliance manual 1.3 Jute product quality manual 1.4 Maintenance procedure manual 1.5 Signs and symbols, instruction manuals
2. Quality improvement system	2.1 Quality inspection 2.2 Testing 2.3 Quality control 2.4 Quality assurance 2.5 Total Quality Management (TQM)
3. Customer quality requirements	3.1 Performance 3.2 Features 3.3 Reliability 3.4 Conformance 3.5 Aesthetics 3.6 Durability

Curricular Content Guide

1. Underpinning Knowledge	1.1 Themes on various types of RMG manuals 1.2 Schedules, dimensions and specifications contained in sketch 1.3 Units of conversion 1.4 Rules of sketch, drawings and specifications 1.5 Sketch and specifications 1.6 Signs and symbols 1.7 Terms and abbreviations used
2. Underpinning Skills	2.1 Recognising importance of manual 2.2 Selecting appropriate manuals as per sample 2.3 Collecting information from manual as per sample 2.4 Identifying signs and symbols 2.5 Identifying drawings and specifications 2.6 Interpreting schedules, dimensions, drawings and specifications
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Manuals 4.3 Workplace documents, signs and symbols 4.4 Sketch and specifications 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected information from manual as per sample 1.2 Identified relevant drawings and specifications 1.3 identified signs and symbols 1.4 identified terms and abbreviations 1.5 identified sketches and specifications as per sample
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN JUTE SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-MJM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in Jute sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices in Jute</u> manufacturing are explained. 1.2 Key terms and symbols in Jute production processes are identified.

	1.3 <u>Sources of environmental impacts</u> during Jute manufacturing activities are identified. 1.4 Jute manufacturing activities use are explained. 1.5 <u>Ways of mitigating environmental impacts</u> in Jute manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in Jute manufacturing are documented. 2.2 <u>Recyclable and non-recyclable items</u> in Jute production processes are identified. 2.3 Procedures to reduce resource consumption in Jute manufacturing are implemented. 2.4 Sustainable alternatives in Jute manufacturing are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste in Jute manufacturing</u> are identified. 3.2 Hazardous waste in jute production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range (May include but not limited to):
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in Jute manufacturing processes. 1.2 6R for waste management in Jute production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in Jute manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in Jute production with low environmental impact. 1.4 Recycling of materials used in the Jute manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the Jute sector.
2. Sources of environmental impacts	2.1 Air pollution in Jute manufacturing 2.2 Water pollution in Jute manufacturing 2.3 Soil degradation in Jute manufacturing

	2.4 Noise pollution in Jute manufacturing 2.5 Waste generation in Jute manufacturing 2.6 Resource depletion in Jute manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in Jute production processes. 3.2 Adopting renewable energy sources for Jute manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 Recyclable Items in RMG Manufacturing 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 Non-Recyclable Items in Jute Manufacturing 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.
5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Jute manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in Jute manufacturing
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	1.6 Hazardous waste in Jute production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in Jute manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in Jute manufacturing 2.6 Disposing hazardous waste in Jute production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation Specific Competencies (180 Hrs.)

Unit of Competency: INTERPRET JUTE MILLS MACHINERY	Nominal Duration: 30 Hrs.	Unit Code: SICIP-JUTE-MJM-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to Interpret Jute Mills Machinery. It specifically includes the tasks of Identifying jute mill machinery and their functions, explaining the operating principles of jute mill machinery, identifying major machine components and their functions, interpreting machine operating parameters and performance requirements, identifying safety requirements and operational hazards associated with jute mill machinery.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify jute mill machinery and their functions	1.1 <u>Jute mill machineries</u> are identified according to their function. 1.2 Machine operating principles are interpreted according to manufacturer instructions. 1.3 Safety requirements are identified according to workplace procedures. 1.4 Machine information is confirmed before maintenance work begins.
2. Explain the operating principles of jute mill machinery.	2.1 Machine operating principles are explained according to manufacturer instructions. 2.2 Material flow through the machine is described correctly. 2.3 Functions of major machine components are explained. 2.4 Machine operating sequence is identified according to work requirements. 2.5 Normal operating parameters are identified according to specifications. 2.6 Operating abnormalities are identified and reported according to workplace procedures.
3. Identify major machine components and their functions.	3.1 Major machine components are identified according to machine manuals. 3.2 Functions of machine components are identified correctly. 3.3 Component locations are identified according to machine layout. 3.4 Component names and specifications are verified. 3.5 Worn or damaged components are identified. 3.6 Component information is recorded according to workplace procedures.
4. Interpret machine	4.1 <u>Machine operating parameters</u> are identified according to specifications.

operating parameters and performance requirements	4.2 Performance requirements are interpreted according to machine manuals. 4.3 Standard operating values are verified against specifications. 4.4 Operating deviations are identified and reported. 4.5 Performance requirements are confirmed before maintenance work begins. 4.6 Machine information is recorded according to workplace procedures.
5. . Identify safety requirements and operational hazards associated with jute mill machinery.	5.1 Safety requirements are identified according to workplace procedures. 5.2 Operational hazards are identified before work begins. 5.3 Required personal protective equipment (PPE) is identified according to safety requirements. 5.4 Safety signs and warning labels are identified correctly. 5.5 Risk control measures are identified according to safety procedures. 5.6 Hazards and unsafe conditions are reported according to workplace procedures.

Range of Variables

Variable	Range (May include but not limited to):
1. Jute mill machineries	1.1 Drawing frame 1.2 Spinning frame 1.3 Winding machine 1.4 Beaming/warping machine 1.5 Weaving loom 1.6 Calendaring machine
2. Machine operating parameters	2.1 Speed 2.2 Draft ratio 2.3 Roller/gauge setting 2.4 Tension 2.5 Pressure 2.6 Temperature (as applicable to machine type)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concepts of jute mill machinery and their functions. 1.2 Principles of jute mill machinery operation. 1.3 Understanding of material flow through jute mill machinery. 1.4 Functions of major machine components. 1.5 Identification of machine components and specifications. 1.6 Concepts of machine operating parameters and performance requirements. 1.7 Methods of identifying machine faults and worn components. 1.8 Requirements for interpreting machine manuals and technical documents. 1.9 Safety requirements for jute mill machinery, including PPE and hazard control. 1.10 Procedures for recording machine information and maintenance records.
2. Underpinning Skills	2.1 Identifying jute mill machinery and their functions. 2.2 Interpreting machine operating principles and material flow. 2.3 Identifying major machine components and their functions. 2.4 Interpreting machine operating parameters and specifications. 2.5 Inspecting machine components for wear and damage. 2.6 Identifying safety requirements and workplace hazards. 2.7 Using appropriate personal protective equipment (PPE). 2.8 Reporting machine faults and unsafe conditions. 2.9 Recording machine and maintenance information. 2.10 Following workplace procedures and manufacturer instructions.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1.1 Workplace (simulated or actual) 1.2 Jute processing machine (actual or simulated) requiring preventive maintenance

	1.3 Standard operating procedure 1.4 Tools, equipment and facilities appropriate to the process or activities. 4.1 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified jute mill machinery and their functions. 1.2 explained the operating principles of jute mill machinery. 1.3 identified major machine components and their functions. 1.4 interpreting machine operating parameters and performance requirements. 1.5 identified safety requirements and operational hazards associated with jute mill machinery.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PREVENTIVE MAINTENANCE OF JUTE MILLS MACHINERY	Nominal Duration: 30 Hrs.	Unit Code: SICIP-JUTE-MJM-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform preventive maintenance of jute mills machinery. It specifically includes task of planning and preparing for preventive maintenance, carrying out routine inspection of machinery, performing cleaning, oiling, greasing and lubrication of machine parts, performing minor adjustment and tightening, and completing and reporting maintenance activities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan and prepare for preventive maintenance	1.1 Maintenance schedule and job order are reviewed to confirm <u>machines</u> , scope and time allocated for the task. 1.2 Required <u>tools & equipment</u> , spare parts and <u>lubricants</u> are selected and checked according to job requirement. 1.3 Personal Protective Equipment (PPE) is worn according to workplace OHS procedures. 1.4 Machine is isolated and locked out/tagged out (LOTO) following safety procedures before maintenance commences. 1.5 Work area is prepared and safety barriers/signage are set up to prevent unauthorized access.
2. Carry out routine inspection of machinery	2.1 <u>Machine components</u> are visually and physically inspected for wear, damage or misalignment. 2.2 Abnormal noise, vibration, temperature and odor are checked and recorded against standard operating parameters. 2.3 Inspection findings are recorded on the maintenance checklist/log sheet according to company format. 2.4 Defects beyond routine maintenance scope are identified and reported to the supervisor/maintenance-in-charge.
3. Perform cleaning, oiling, greasing and lubrication of machine parts	3.1 Machine surfaces, guards and moving parts are cleaned of jute dust, fibre waste and debris using appropriate methods and materials. 3.2 Lubrication points are identified according to the manufacturer's lubrication chart. 3.3 Correct type and quantity of lubricant (grease/oil) is applied to designated points according to schedule. 3.4 Excess lubricant and waste materials are removed and

	disposed of according to workplace environmental procedures.
4. Perform minor adjustment and tightening	4.1 Loose fasteners, belts, chains and couplings are identified and tightened/adjusted to manufacturer's tolerance. 4.2 Belt tension, roller setting and guide alignment are adjusted within specified tolerance. 4.3 <u>Safety guards</u> and covers are refitted and secured after adjustment. 4.4 Machine is test-run to confirm smooth operation after adjustment.
5. Complete and report maintenance activities	5.1 Maintenance actions performed are recorded in the maintenance register/CMMS according to company procedure. 5.2 <u>Work area</u> is cleaned and tools/equipment are returned to designated storage. 5.3 Machine is handed over to production with confirmation of safe operating condition. 5.4 Maintenance report is submitted to the supervisor within the specified timeframe.

Range of Variables

Variable	Range (May include but not limited to):
1. Machines	1.1 Emulsion plant 1.2 Softener 1.3 Sprader 1.4 Breaker card and finisher card 1.5 Drawing frame 1.6 Spinning frame 1.7 Winding machine 1.8 Beaming/warping machine 1.9 Weaving loom (Hessian/sacking/CBC) 1.10 Damping 1.11 Calendaring machine 1.12 Lapping 1.13 Cutting 1.14 Sewing
2. Tools and equipment	2.1 Hammer 2.2 Spanners/wrenches 2.3 Screwdrivers 2.4 Allen keys 2.5 Pliers 2.6 Scissor 2.7 Grease gun 2.8 Oil can 2.9 Feeler gauge

	2.10 Hand grinder machine 2.11 Cleaning brush/air blower
3. Lubricants	3.1 Lithium-based grease 3.2 Gear oil 3.3 Spindle oil
4. Machine components	4.1 Staves 4.2 Rollers 4.3 Gears 4.4 Belts 4.5 Bearings 4.6 Chains
5. Safety guard	5.1 Lockout/tagout (LOTO) procedure 5.2 PPE (apron, gloves, goggles, ear muffs, safety shoes, hair net/cap) 5.3 Safety signage and barricading 5.4 Machine guarding
6. Maintenance register	6.1 Schedules of preventive maintenance 6.2 Lubrication chart 6.3 Checklist/log sheet 6.4 Job card
7. Work area	7.1 Production floor 7.2 Workshop 7.3 Corridor

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and functions of jute mill machinery. 1.2 Principles of preventive maintenance. 1.3 Types and application of lubricants. 1.4 Reading and interpreting maintenance manuals and job cards. 1.5 Lockout/Tagout (LOTO) and machine isolation procedures. 1.6 Functions of basic mechanical components. 1.7 Common causes of machine wear and damage. 1.8 Housekeeping and dust control practices. 1.9 OHS requirements for machine maintenance. 1.10 Procedures for maintenance records and reporting.
2. Underpinning Skills	2.1 Reading maintenance schedules and lubrication charts 2.2 Selecting correct tools, lubricants and PPE for the task 2.3 Isolating and locking out machinery safely 2.4 Conducting visual and physical inspection of machine parts 2.5 Identifying abnormal noise, vibration and temperature 2.6 Cleaning machinery and work area using correct materials/methods 2.7 Applying lubrication correctly to designated points 2.8 Adjusting and tightening fasteners, belts and guides within tolerance

	2.9 Recording maintenance data accurately in logs/CMMS 2.10 Communicating maintenance findings to supervisors
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Jute processing machine (actual or simulated) requiring preventive maintenance 4.3 Standard operating procedure 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 planed and prepared for preventive maintenance 1.2 carried out routine inspection of machinery 1.3 performed cleaning. oiling, greasing and lubrication of machine parts 1.4 performed minor adjustment and tightening 1.5 completed and report maintenance activities
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DIAGNOSE MECHANICAL FAULTS IN JUTE MACHINERY	Nominal Duration: 35 Hrs.	Unit Code: SICIP-JUTE-MJM-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to diagnose mechanical faults in jute machinery. It specifically includes task of Identifying symptoms of machine malfunction using diagnostic tools and techniques, analyzing root cause of faults, determining and recommend corrective action and documenting fault diagnosis findings		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify symptoms of machine malfunction	1.1 Machine performance/output is monitored against standard operating parameters to detect abnormalities. 1.2 Unusual sound, vibration, heat, smell or output quality defects are observed and recorded. 1.3 Operator/production reports on machine performance are reviewed to identify potential fault indicators. 1.4 <u>Fault symptoms</u> are correlated with possible machine sub-systems (mechanical drive) for further diagnosis.
2. Use diagnostic tools and techniques	2.1 Appropriate <u>diagnostic instruments</u> are selected according to the suspected fault type. 2.2 Diagnostic tests are carried out following manufacturer's procedures and safety precautions. 2.3 Test readings are compared with standard/benchmark values to confirm deviation.
3. Analyze root cause of faults	3.1 Fault data and inspection findings are analyzed using systematic <u>fault-finding techniques</u> 3.2 <u>Mechanical causes</u> are distinguished from electrical/drive-related causes. 3.3 Root cause of the fault is identified and validated against machine history/records.
4. Determine and recommend corrective action	4.1 Appropriate <u>corrective actions</u> are determined based on root cause analysis. 4.2 Required spare parts, tools and time needed for corrective action are estimated. 4.3 Priority and urgency of corrective action are assessed based on production impact and safety risk.
5. Document fault diagnosis findings	5.1 Fault diagnosis details, root cause and recommended action are recorded in the maintenance/fault report according to company format. 5.2 Machine history record/CMMS is updated with diagnosis and outcome.

	5.3 Recurring fault patterns are flagged for further investigation or design review.
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Range of Variables

Variable	Range (May include but not limited to):
1. Fault symptoms	1.1 Mechanical faults (wear, misalignment, breakage) 1.2 Drive-related faults (belt slip, chain wear, gearbox fault) 1.3 Pneumatic/hydraulic faults (where applicable)
2. Diagnostic instruments	2.1 Tachometer 2.2 Vibration meter 2.3 Thermal/infrared gun 2.4 Feeler gauge 2.5 Dial gauge 2.6 Mechanic's stethoscope 2.7 Vernier Scale 2.8 Micrometer
3. fault-finding techniques	3.1 Process of elimination 3.2 Cause-and-effect analysis
4. Mechanical causes	4.1 Wear 4.2 Misalignment 4.3 Lubrication failure
5. Corrective action	5.1 Repair, 5.2 Replacement 5.3 Adjustment

Curricular Content Guide

1. Underpinning Knowledge	1.1 Working principles of jute machinery (mechanical, drive and electrical systems) 1.2 Common fault types and their symptoms in jute machinery 1.3 Function and use of diagnostic instruments (multimeter, megger, vibration meter, tachometer, thermal gun) 1.4 Basic electrical concepts - voltage, current, resistance, insulation testing 1.5 Fault-finding techniques and root cause analysis methods 1.6 Reading wiring diagrams and machine manuals 1.7 Standard operating parameters/tolerances for jute machinery 1.8 Machine history recording and CMMS usage 1.9 Electrical safety and isolation procedures 1.2 Escalation procedures for complex or safety-critical faults
2. Underpinning Skills	2.1 Monitoring machine performance to detect abnormalities 2.2 Operating diagnostic instruments safely and accurately

	2.3 Interpreting diagnostic readings against standard values 2.4 Applying systematic fault-finding/root cause analysis techniques 2.5 Distinguishing mechanical faults from electrical/drive faults 2.6 Estimating spare parts, tools and time for corrective action 2.7 Prioritizing corrective actions based on risk and production impact 2.8 Documenting fault diagnosis and outcomes accurately 2.9 Communicating technical findings to supervisors/team 2.10 Reading and interpreting wiring/circuit diagrams
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Jute processing machinery with induced or actual faults for diagnostic practice 4. 3 Diagnostic instruments (multimeter, megger, vibration meter, tachometer, thermal gun) 4. 4 Rules and regulations 4. 5 Internet connectivity 4. 6 Tools, equipment and facilities appropriate to the process or activities.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for repair/replacement work 1.2 dismantled machine components 1.3 repaired or replaced defective components 1.4 reassembled and reinstall components 1.5 tested and commissioned repaired machine
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of	4.6 Competency assessment must be done in an

Assessment	assessment/training center or in an actual or simulated work place after completion of the training. 3.3 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: CARRY OUT REPAIR AND REPLACEMENT OF JUTE MACHINERY COMPONENTS	Nominal Duration: 35 Hrs.	Unit Code: SICIP-JUTE-MJM-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out repair and replacement of jute machinery components. It specifically includes task of preparing for repair/replacement work, dismantling machine components, repairing or replace defective components, reassembling and reinstall components and testing and commissioning repaired machine.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for repair/replacement work	1.1 Job order/work instruction is reviewed to confirm scope, components and spare parts required. 1.2 Required tools, lifting equipment and spare parts are arranged and checked for availability and condition. 1.3 Machine is isolated, de-energized and locked out/tagged out according to safety procedures. 1.4 PPE and safety barriers are used/set up appropriate to the repair task.
2. Dismantle machine components	2.1 Covers, guards and connecting parts are removed in correct sequence using appropriate tools. 2.2 Components are dismantled using correct technique to avoid damage to adjoining parts. 2.3 Removed parts, fasteners and small components are labeled/organized to facilitate reassembly. 2.4 Condition of removed components is inspected and compared against tolerance/wear limits.
3. Repair or replace defective components	3.1 <u>Defective components</u> are repaired or replaced according to manufacturer's specification. 3.2 Correct replacement parts are selected matching specification, size and grade. 3.3 <u>Repair techniques</u> (welding, machining, resurfacing where applicable) are applied according to standard procedure and within scope of competency. 3.4 Repaired/replaced components are checked for

	dimensional accuracy and fit before reassembly.
4. Reassemble and reinstall components	4.1 Components are reassembled in correct sequence following manufacturer's assembly procedure. 4.2 Fasteners are tightened to specified torque and alignment tolerance. 4.3 Belts, chains and drives are correctly tensioned and aligned. 4.4 Safety guards and covers are refitted and secured.
5. Test and commission repaired machine	5.1 Machine is test-run under no-load and then load condition to check performance. 5.2 Abnormal noise, vibration or temperature is checked and corrected before handover. 5.3 Repair/replacement details are recorded in maintenance history/CMMS. 5.4 Machine is handed over to production with confirmation of safe and normal operating condition.

Range of Variables

Variable	Range (May include but not limited to):
1. Defective components	1.1 Bearings, 1.2 Belts 1.3 Gears 1.4 Rollers 1.5 Chains 1.6 Card clothing 1.7 Bushings
2. Repair techniques	2.1 Replacement 2.2 Resurfacing 2.3 Machining (within scope) 2.4 Welding/brazing (within scope) 2.5 Balancing

Curricular Content Guide

1. Underpinning Knowledge	1.1 Function and construction of major jute machinery components (bearings, gears, belts, rollers, chains, spindles, clothing) 1.2 Wear limits and tolerances for mechanical components 1.3 Correct dismantling and reassembly sequence and techniques 1.4 Selection criteria for replacement spare parts 1.5 Basic repair techniques (resurfacing, machining, welding) relevant to maintenance scope 1.6 Torque specifications and fastening standards 1.7 Belt/chain tensioning and alignment principles
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	1.8 Safe lifting and material handling practices 1.9 LOTO and machine isolation procedures for repair work
2. Underpinning Skills	2.1 Interpreting job orders and technical drawings/specifications 2.2 Selecting and using hand tools, power tools and lifting equipment safely 2.3 Dismantling and labeling machine components systematically 2.4 Inspecting components against wear/tolerance limits 2.5 Repairing or replacing defective mechanical components 2.6 Reassembling components to manufacturer's specification 2.7 Aligning and tensioning belts, chains and drives 2.8 Conducting test-run and verifying machine performance 2.9 Recording repair/replacement activities in maintenance records 2.10 Applying safe work practices during repair activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Attendance and performance record templates 4.3 Grievance handling procedure documents 4.4 Bangladesh Labor Act reference materials 4.5 Case study scenarios for grievance handling 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for repair/replacement work 1.2 dismantled machine components 1.3 repaired or replaced defective components 1.4 reassemble and reinstall components 1.5 tested and commission repaired machine
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test

	2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM MACHINE TESTING AND COMMISSIONING	Nominal Duration: 25 Hrs.	Unit Code: SICIP-JUTE-MJM-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform machine testing and commissioning. It specifically includes task of preparing machine for alignment and calibration performing mechanical alignment, calibrating machine settings and commission and validate machine performance		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare machine for alignment and calibration	1.1 Maintenance/repair completion status is confirmed prior to alignment/calibration. 1.2 Required <u>alignment and calibration tools/instruments</u> are selected and checked for accuracy/calibration validity. 1.3 Machine is positioned, leveled and secured for alignment work. 1.4 Reference standards/specifications for alignment and calibration are obtained from the machine manual.
2. Perform mechanical alignment	2.1 <u>Machine alignments</u> are checked using dial indicator/laser alignment tool. 2.2 Misalignment is corrected within manufacturer's tolerance. 2.3 Belt/chain tension and guide settings are adjusted to specification after alignment. 2.4 Alignment results are verified and recorded.
3. Calibrate machine settings	3.1 <u>Machine settings</u> are calibrated according to product specification. 3.2 Calibration instruments/gauges are used correctly to verify settings against standard values. 3.3 Deviations from standard settings are adjusted and re-

	verified. 3.4 Calibration results are documented according to company/quality procedure.
4. Commission and validate machine performance	4.1 Machine is run under trial/production condition to validate performance against required output and quality standard. 4.2 Product/output sample is checked against quality specification and process parameters. 4.3 Final adjustments are made based on trial run results. 4.4 Commissioning results are recorded and the machine is formally handed over to production with sign-off.

Range of Variables

Variable	Range (May include but not limited to):
1. alignment and calibration tools/instruments	1.1 Dial indicator 1.2 Laser alignment tool 1.3 Straight edge 1.4 Feeler gauge 1.5 Vernier caliper 1.6 Micrometer 1.7 Tachometer
2. Machine alignments	2.1 Shaft 2.2 Pulley 2.3 Roller and 2.4 Drive
3. Machine settings	2. 1 Speed 2. 2 Draft ratio 2. 3 Roller/gauge setting 2. 4 Tension 2. 5 Pressure gauge 2. 6 Temperature (as applicable to machine type)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of mechanical alignment (shaft, pulley, belt/chain alignment) 1.2 Use of alignment tools - dial indicator, laser alignment device, straight edge 1.3 Machine-specific operating parameters and tolerances (speed, draft, tension, gauge) 1.4 Calibration principles and use of measuring instruments 1.5 Product quality specifications relevant to jute processing/weaving 1.6 Trial-run and commissioning procedures 1.7 Reading product specification sheets and quality checklists 1.8 Documentation and sign-off procedures for
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	commissioning 1.9 Impact of misalignment/miscalibration on product quality and machine life
2. Underpinning Skills	2.1 Selecting and using alignment and calibration instruments correctly 2.2 Checking and correcting shaft/pulley/belt alignment 2.3 Calibrating machine settings against product specification 2.4 Conducting trial runs and evaluating output quality 2.5 Making fine adjustments based on trial run results 2.6 Interpreting product specification and quality standards 2.7 Recording calibration and commissioning results 2.8 Communicating commissioning outcome and obtaining sign-off
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Tools, equipment and facilities appropriate to the process or activities. 4. 3 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. 1 prepared machine for alignment and calibration 1. 2 performed mechanical alignment 1. 3 calibrated machine settings 1. 4 commissioned and validated machine performance
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN JUTE PROCESSING MACHINES	Nominal Duration: 25 Hrs.	Unit Code: SICIP-JUTE-MJM-06-0
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to maintain jute processing machines. It specifically includes task of maintaining emulsion plan and softening machines, maintaining breaker card and finisher card machines, maintaining drawing frame and spinning frame machines, maintaining winding, beaming, and weaving machines, maintaining finishing and baling equipment and verifying machine performance and production readiness after maintenance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Maintain emulsion plan and softening machines.	1.1 Machine condition is inspected according to workplace procedures. 1.2 Maintenance requirements are identified according to maintenance schedules. 1.3 Machine components are cleaned according to manufacturer instructions. 1.4 Worn or damaged parts are identified and replaced. 1.5 <u>Machine adjustments</u> are carried out according to specifications. 1.6 Machine performance is checked after maintenance. 1.7 Maintenance activities are recorded according to workplace procedures.
2. Maintain breaker card and finisher card machines	2.1 Maintenance requirements are identified according to maintenance schedules. 2.2 Machine condition is inspected according to workplace procedures. 2.3 Machine components are cleaned according to manufacturer instructions. 2.4 Worn or damaged parts are identified and replaced. 2.5 Machine adjustments are carried out according to specifications. 2.6 Machine performance is checked after maintenance. 2.7 <u>Maintenance activities</u> are recorded according to workplace procedures.
3. Maintain drawing frame and spinning frame machines.	3.1 Maintenance requirements are identified according to maintenance schedules. 3.2 Machine condition is inspected according to workplace

	procedures. 3.3 Machine components are cleaned according to manufacturer instructions. 3.4 Worn or damaged parts are identified and replaced. 3.5 Machine adjustments are carried out according to specifications. 3.6 Machine performance is checked after maintenance. 3.7 Maintenance activities are recorded according to workplace procedures.
4. Maintain winding, beaming, and weaving machines.	4.1 Maintenance requirements are identified according to maintenance schedules. 4.2 Machine condition is inspected according to workplace procedures. 4.3 Machine components are cleaned according to manufacturer instructions. 4.4 Worn or damaged parts are identified and replaced. 4.5 Machine adjustments are carried out according to specifications. 4.6 Machine performance is checked after maintenance. 4.7 Maintenance activities are recorded according to workplace procedures.
5. Maintain finishing and baling equipment.	5.1 Maintenance requirements are identified according to maintenance schedules. 5.2 Equipment condition is inspected according to workplace procedures. 5.3 Equipment components are cleaned according to manufacturer instructions. 5.4 Worn or damaged parts are identified and replaced. 5.5 Equipment adjustments are carried out according to specifications. 5.6 Equipment performance is checked after maintenance.
6. Verify machine performance and production readiness after maintenance.	6.1 Machine operation is tested according to manufacturer specifications. 6.2 Machine performance is verified against operating requirements. 6.3 Machine adjustments are confirmed before production. 6.4 Safety devices are checked for proper operation. 6.5 Production readiness is verified according to workplace procedures. 6.6 Maintenance records are completed according to workplace procedures.

Range of Variables

Variable	Range
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	(May include but not limited to):
1. Machine adjustments	1.1 Alignment 1.2 tension 1.3 clearance 1.4 calibration 1.5 timing 1.6 lubrication
2. Maintenance activities	1.7 Inspection 1.8 cleaning 1.9 lubrication 1.10 adjustment 1.11 replacement 1.12 testing 1.13 recording
3. Safety devices	1.14 Machine guards 1.15 emergency stop 1.16 interlocks 1.17 safety sensors
4. Maintenance records	1.18 Maintenance logbook 1.19 inspection report 1.20 job card 1.21 service record

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concepts of jute mill machine maintenance. 1.2 Procedures for machine inspection and maintenance. 1.3 Methods of cleaning machine components. 1.4 Identification of worn and damaged machine parts. 1.5 Procedures for machine adjustment. 1.6 Principles of machine operation and performance. 1.7 Methods of checking machine performance. 1.8 Requirements for machine testing and production readiness. 1.9 Safety requirements for machine maintenance. 1.10 Procedures for maintenance records and documentation.
2. Underpinning Skills	2.1 Applying maintenance schedules to plan machine servicing activities. 2.2 Inspecting machine and equipment conditions to identify maintenance requirements. 2.3 Cleaning machine components using manufacturer-recommended procedures.

	2.4 Identifying and replacing worn or damaged machine parts. 2.5 Performing machine adjustments according to technical specifications. 2.6 Using appropriate hand tools and maintenance equipment safely and effectively. 2.7 Testing machine performance after maintenance to verify proper operation. 2.8 Checking safety devices and machine guards before production. 2.9 Verifying machine and equipment readiness for production according to workplace procedures. 2.10 Recording maintenance activities and inspection results according to workplace procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Writing materials. 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. 1 maintained batching and softening machines. 1. 2 maintained breaker card and finisher card machines 1. 3 maintained drawing frame and spinning frame machines. 1. 4 maintained winding, beaming, and weaving machines. 1. 5 maintained finishing and baling equipment. 1. 6 verifying machine performance and production readiness after maintenance.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Maintenance of Jute Mills Machinery
Number of Trainees:	30

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and Tools	Required Facilities
1. 1	Desktop Computers	1 Nos.
2. 2	Instructor Computer/Laptop	1 Nos
3. 3	Multimedia Projector	1 Nos
4. 4	Whiteboard and markers	1 Nos
5. 5	Flip chart stands with paper	1 Nos
6. 6	Printer	1 Nos
7. 7	Scanner	1 Nos
8. 8	Dust shaker machine	1 Nos
9.	Emulsion plant	1 Nos
10.	Softener	1 Nos
11.	Spaeder	1 Nos
12. 9	Breaker Card machine	1 Nos
13. 1	Finisher Card machine	1 Nos
14. 1	Drawing Machine	1 Nos
15. 1	Spinning Machine	1 Nos

16. 1	Winding Machine	1 Nos
17. 1	Beaming	1 Nos
18. 1	Weaving Machine	1 Nos
19. 1	Inspection machine	1 Nos
20. 1	Lapping machine	1 Nos
21. 1	Rolling Up	1 Nos
22. 1	Cutting Machine	1 Nos
23. 2	Sewing Machine (Hemming-01, Herackle-01, Safety Sewing-01, Over Head Dry/ Tar Sewing)	1 Nos
24. 2	Measuring Steel Tape	1 Nos
25. 2	Moisture Meter	1 Nos
26. 2	Tucking device	1 Nos
27. 2	Loom Dropper	1 Nos
28. 2	Heald Wires	1 coil.
29. 2	Mechanical kit box	1 Nos.
30. 3	Digital scale	1 Nos
31. 3	First aid kit	1 Nos
32. 3	Fire extinguisher (for safety demonstration)	2 Nos

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The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Maintenance of Jute Mills Machinery (Jute sector)** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 30 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 30 trainees.